

Business Requirements Document

Transaction Anomaly Review System

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1. BUSINESS OBJECTIVE

Reduce the platform's exposure to fraudulent, erroneous, or otherwise unusual financial transactions by identifying and prioritizing high-risk transactions for manual review, without materially increasing operational review workload or introducing friction for legitimate customers.

2. BACKGROUND

The platform processes digital payments across multiple modes (UPI, cards, net banking, wallets). Currently, unusual transactions are only caught reactively — typically after a customer complaint or a chargeback — which delays response and increases financial and reputational risk. There is no systematic, proactive mechanism to flag transactions that deviate from a customer's normal behavior at the time they occur.

3. STAKEHOLDERS

- Risk & Fraud Operations team — primary users of the review queue
- Customer Support — handles downstream customer queries on flagged transactions
- Product / Engineering — owns integration of scoring into the transaction pipeline
- Compliance — ensures review process meets regulatory expectations for transaction monitoring

4. BUSINESS REQUIREMENTS

- BR-1: The system shall evaluate every transaction against the originating customer's own historical behavior (typical amount, typical location, typical time of day).
- BR-2: The system shall assign a review-priority score to each transaction, combining rule-based behavioral signals with an independent statistical/ML anomaly check.
- BR-3: Transactions exceeding a defined review-score threshold shall be routed to a prioritized queue for the Risk Operations team, rather than surfaced as raw alerts.
- BR-4: The flag rate shall remain low enough (target: under 3% of volume) to keep the review queue actionable rather than overwhelming.
- BR-5: Reviewers shall be able to see the specific reason(s) a transaction was flagged (e.g., unusual amount, off-home-city, odd hour), not just a binary flag.
- BR-6: Review outcomes (confirmed issue / cleared as legitimate) shall be logged to support future recalibration of thresholds.

5. PROPOSED APPROACH

A per-customer behavioral baseline is established from historical transactions (average amount, typical city, typical hours). Each new transaction is scored against this baseline using: (a) an amount deviation (z-score) check, (b) an off-home-city flag, (c) an odd-hour flag, and (d) an independent Isolation Forest model as a cross-check against the rule-based score. Transactions meeting the combined threshold are routed to a review queue, visualized in a dashboard giving Operations a prioritized, explainable view rather than a raw data dump.

6. SUCCESS METRICS

- Review queue captures high-value anomalies while keeping flag volume under ~3% of total transactions (pilot result: 2.16%)
- Time-to-detection for unusual transactions reduced from post-complaint (reactive) to same-day (proactive)
- Reviewer feedback confirms flagged reasons are clear and actionable, reducing average review time per case

7. OUT OF SCOPE (THIS PHASE)

- Automated transaction blocking or holds — this phase is detection and review-routing only, not enforcement

- Real-time streaming scoring — pilot operates on batch-scored historical data; real-time integration is a future phase

8. ASSUMPTIONS & CONSTRAINTS

- Pilot analysis uses a representative synthetic dataset (800 customers, ~77,000 transactions over 6 months) standing in for production transaction data
- Thresholds (z-score, review-score cutoff) are tunable and expected to be recalibrated after live review feedback